Specialization Elective Project

Comparative Study of Models for Transmission of Squeezed Light in Atmospheric Channels

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1 Introduction

Quantum communication is an emerging field that uses the principles of quantum mechanics to transmit information securely. Light, especially in the form of quantum states, plays a crucial role as an information carrier in such systems. Among these states, quadrature-squeezed light is particularly important because it allows reduction of noise below classical limits, making it useful for precision measurements and secure communication. However, when light propagates through the atmosphere, it is affected by turbulence. Fluctuations in the refractive index of air lead to effects such as beam wandering, spreading, and intensity fluctuations. These disturbances result in random losses in the transmitted signal, which can degrade the nonclassical properties of squeezed light. To understand these effects, atmospheric channels are often described using the probability distribution of transmittance (PDT), which characterizes how much of the light is transmitted through the channel under varying conditions. Several theoretical models, such as the log-normal model, beam-wandering model, and Beta distribution model, have been proposed to describe these fluctuations. In this report, we study and compare different models used to describe atmospheric quantum channels and analyze their impact on the transmission of quadrature-squeezed light.

2 Theoretical Background

This section presents the essential theoretical concepts underlying the study of quantum light propagation in atmospheric channels. We review quantum states of light, with emphasis on quadrature squeezing, and discuss the impact of atmospheric turbulence on optical beams. The concept of channel transmittance and its statistical nature are also introduced, providing the foundation for the models analyzed later.

2.1 Quantum States of Light

In quantum optics, light is described using quantum states of the electromagnetic field. A commonly used state is the coherent state, which closely resembles classical light and has equal uncertainty in all field quadratures.

Nonclassical states, such as squeezed states, exhibit reduced quantum noise in one observable at the expense of increased noise in its conjugate variable. These states are particularly important in quantum communication and precision measurements.

2.2 Quadrature Squeezing

The properties of quantum light are described in terms of field quadratures. One such quadrature operator is given by:

$$\hat{x} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger) \tag{1}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operators.

For a coherent or vacuum state, the quadrature variance is:

$$\langle(\Delta\hat{x})^2\rangle = \frac{1}{2} \tag{2}$$

A state is said to be squeezed if:

$$\langle(\Delta\hat{x})^2\rangle < \frac{1}{2} \tag{3}$$

This reduction below the standard quantum limit indicates nonclassical behavior and is the defining feature of quadrature squeezing.

2.3 Atmospheric Effects on Light Propagation

When light propagates through the atmosphere, turbulence causes fluctuations in the refractive index. The evolution of the optical field amplitude $u(\mathbf{r}, z)$ is described by the paraxial wave equation:

$$2ik\frac{\partial u}{\partial z} + \nabla^2 u + 2k^2\delta n(\mathbf{r}, z)\, u = 0 \tag{4}$$

Here, k is the wave number and $\delta n(\mathbf{r}, z)$ represents fluctuations in the refractive index. These fluctuations lead to beam wandering, beam broadening, and intensity fluctuations, which distort the beam and introduce randomness in transmission.

2.4 Channel Transmittance

To quantify losses, the transmittance η is defined as the fraction of optical power received:

$$\eta = \int_A I(\mathbf{r}, z)\, d^2\mathbf{r} \tag{5}$$

where $I(\mathbf{r}, z) = |u(\mathbf{r}, z)|^2$ is the intensity and A is the receiver aperture. Due to atmospheric turbulence, η becomes a random variable. Its statistical behavior is described by the probability distribution of transmittance (PDT).

2.5 Input–Output Relation

The effect of fluctuating losses on quantum states can be described using the input–output relation:

$$P_{out}(\alpha) = \int_0^1 \frac{d\eta}{\eta}\, P_{in}\left(\frac{\alpha}{\sqrt{\eta}}\right) P(\eta) \tag{6}$$

where P_{in} and P_{out} are the input and output state distributions, and $P(\eta)$ is the probability distribution of transmittance. This relation shows how atmospheric fluctuations modify the quantum state of light.

2.6 Overview of Modeling Approach

Since transmittance fluctuates randomly, different models are used to describe its probability distribution. These models aim to capture the effects of atmospheric turbulence and provide a framework for analyzing how quantum states, such as squeezed light, are affected during transmission. In the next section, we will discuss and compare these models in detail.

3 Models of Atmospheric Quantum Channels

In atmospheric quantum channels, fluctuations in transmission are described using the probability distribution of transmittance (PDT). Several models have been proposed to characterize these fluctuations under different physical assumptions. In this section, we discuss the most commonly used models.

3.1 Truncated Log-Normal Model

The log-normal distribution is widely used to describe intensity fluctuations in weak turbulence. Since transmittance is bounded between 0 and 1, a truncated version is used:

$$P(\eta) = \frac{1}{F(1)} \frac{1}{\sqrt{2\pi}\sigma\eta} \exp\left[-\frac{(\ln\eta + \mu)^2}{2\sigma^2}\right] \tag{7}$$

Here, μ and σ are parameters related to the mean and variance of transmittance. This model is simple and effective for weak turbulence conditions.

3.2 Beam-Wandering Model

In this model, turbulence is assumed to mainly cause random displacement of the beam centroid. The beam shape is approximated as Gaussian, and fluctuations arise due to beam wandering.

The beam width is modified as:

$$W_{ST}^2 = W_{LT}^2 - 4\sigma_{bw}^2 \tag{8}$$

where W_{LT} is the long-term beam width and σ_{bw}^2 represents the variance of beam wandering.

This model is suitable when beam displacement dominates over beam distortion.

3.3 Elliptic-Beam Model

The elliptic-beam model accounts for both beam wandering and deformation of the beam shape. The intensity profile is approximated as an elliptic Gaussian:

$$I(\mathbf{r}) = \frac{2}{\pi\sqrt{\det S}} \exp\left[-2(\mathbf{r} - \mathbf{r}_0)^T S^{-1} (\mathbf{r} - \mathbf{r}_0)\right] \tag{9}$$

Here, $\mathbf{r}_0$ represents the beam centroid and S is the spot-shape matrix.

This model provides a more accurate description of turbulence effects but is more complex.

3.4 Model Based on the Law of Total Probability

This model combines the effects of beam wandering and beam distortion using conditional probability:

$$P(\eta) = \int d^2\mathbf{r}_0 \, P(\eta|\mathbf{r}_0) \, \rho(\mathbf{r}_0) \tag{10}$$

where $\rho(\mathbf{r}_0)$ describes the distribution of beam centroid positions and $P(\eta|\mathbf{r}_0)$ is typically modeled using a log-normal distribution.

This approach captures multiple turbulence effects simultaneously.

3.5 Beta-Distribution Model

An alternative empirical model uses the Beta distribution, which is naturally defined in the interval $[0, 1]$:

$$P(\eta) = \frac{1}{B(a,b)} \eta^{a-1} (1-\eta)^{b-1} \tag{11}$$

where $B(a, b)$ is the Beta function, and parameters a and b are determined from the moments of transmittance.

This model is flexible and provides good agreement with numerical simulations over a wide range of conditions.

3.6 Summary of Models

Each model is based on different assumptions about atmospheric effects. The log-normal model is suitable for weak turbulence, while the beam-wandering and elliptic-beam models account for spatial effects. The Beta distribution provides a simple yet effective empirical description. In the next section, we compare these models and analyze their performance.

4 Results

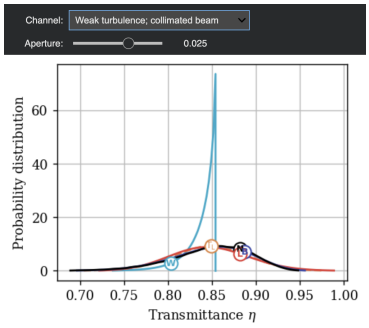
This section presents the probability distribution of transmittance (PDT) for different atmospheric channel conditions. The PDT is evaluated using multiple models, including the truncated log-normal, beam-wandering, elliptic-beam, and Beta-distribution models.

Five representative channel scenarios are considered, corresponding to different turbulence strengths and beam configurations:

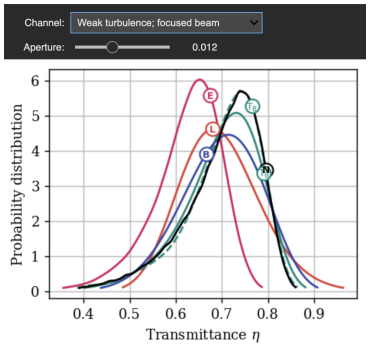
- Weak turbulence, collimated beam
- Weak turbulence, focused beam
- Moderate turbulence, collimated beam
- Moderate turbulence, focused beam
- Strong turbulence, collimated beam

For each scenario, the corresponding PDT curves obtained from different models are plotted and compared. The figures display the variation in the predicted distributions under different channel conditions, with all models shown on the same axes for direct comparison.

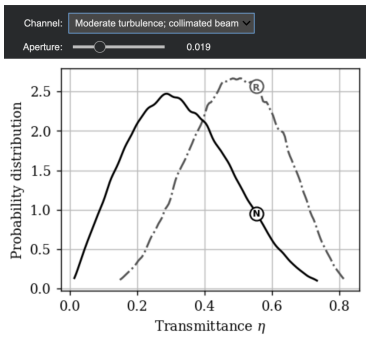
The parameters used for each case, including aperture size and channel characteristics, are specified in the respective plots.



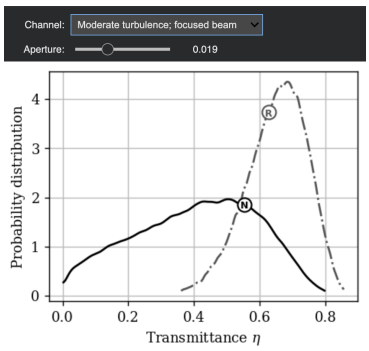
(a) Weak turbulence, collimated beam
($R_{ap} = 0.025$)



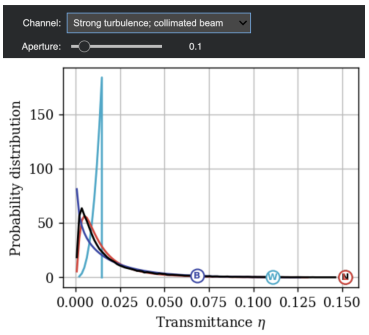
(b) Weak turbulence, focused beam
($R_{ap} = 0.012$)



(c) Moderate turbulence, collimated beam
($R_{ap} = 0.019$)



(d) Moderate turbulence, focused beam
($R_{ap} = 0.019$)



(e) Strong turbulence, collimated beam
($R_{ap} = 0.1$)

Figure 1: Comparison of probability distribution of transmittance (PDT) predicted by different models for various atmospheric channel conditions.

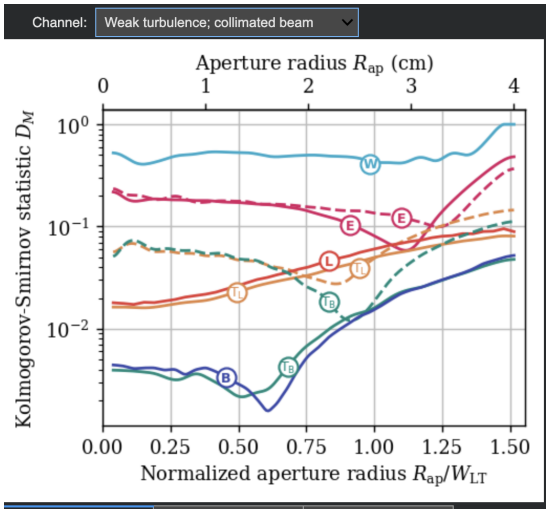
4.1 Kolmogorov–Smirnov (KS) Statistic

To quantitatively compare the agreement between different models, the Kolmogorov–Smirnov (KS) statistic is used. It measures the maximum difference between the cumulative distribution functions (CDFs) of the model and the reference distribution:

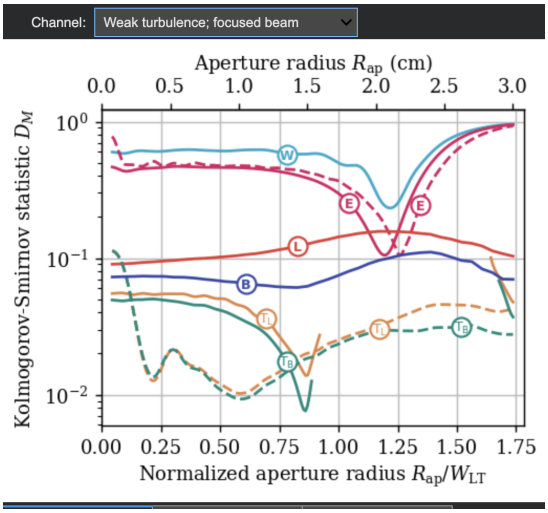
$$D_M = \sup_{\eta} |F_{model}(\eta) - F_{ref}(\eta)| \tag{12}$$

A smaller value of D_M indicates better agreement between the model and the reference data.

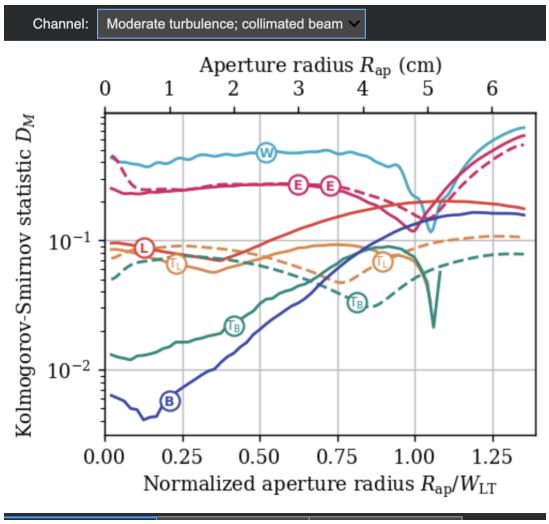
The variation of the KS statistic with aperture radius is shown for different channel conditions. In some cases, the aperture radius is also expressed in normalized form to enable comparison across different beam configurations.



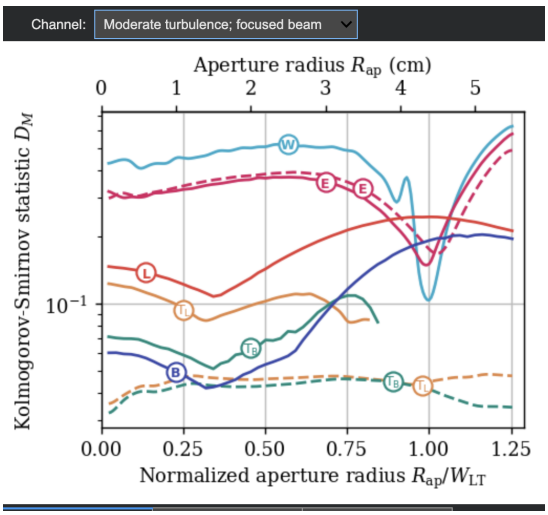
(a) Weak turbulence, collimated beam



(b) Weak turbulence, focused beam



(c) Moderate turbulence, collimated beam



(d) Moderate turbulence, focused beam

Figure 2: KS statistic D_M as a function of aperture radius (or normalized aperture radius) for different atmospheric channel conditions.

5 Analysis

5.1 Behavior of PDT with Turbulence

The PDT changes a lot depending on how strong the turbulence is.

For weak turbulence (Fig. (a) and Fig. (b)), the curves are narrow and mostly centered at higher values of transmittance. This means that most of the light is reaching the receiver and fluctuations are small. In the focused beam case, the curve is even more concentrated, which shows that focusing helps in improving transmission. When moving to moderate turbulence (Fig. (c) and Fig. (d)), the curves start spreading out. The peak shifts towards lower values of transmittance, and the distribution becomes wider. This shows that the channel is becoming more unstable and losses are increasing. In the strong turbulence case (Fig. (e)), the PDT looks very different. There is a sharp peak near very small values of transmittance, and a long tail. This means that most of the time, very

little light is received, and high transmission happens only rarely.

5.2 Effect of Beam Configuration

Comparing focused and collimated beams, it is seen that focusing generally gives a better result, especially in weak turbulence. The PDT is narrower and shifted towards higher transmittance. However, as turbulence increases, this difference becomes smaller. In moderate turbulence, both cases show similar spreading. In strong turbulence, the effect of beam configuration is not very significant compared to the overall distortion caused by the atmosphere.

5.3 Model Comparison

In weak turbulence, all models give similar results, and the difference between them is not very large. But in moderate and strong turbulence, the difference becomes clear. Some models are not able to capture the shape of the PDT properly, especially the skewness and long tail. The Beta model follows the overall shape more closely in most cases. It is able to adjust better to different conditions compared to the other models. Overall, no single model works perfectly in all cases, but the Beta model gives more consistent results across different turbulence regimes.

5.4 KS Test Analysis

The Kolmogorov–Smirnov (KS) statistic is used to check how well each model matches the actual distribution. A smaller KS value means a better fit.

5.4.1 Weak Turbulence

In the weak turbulence case, the KS values are generally low. This means that most models are able to match the PDT reasonably well. The difference between models is not very large, especially at smaller aperture sizes.

5.4.2 Moderate Turbulence

In moderate turbulence, the behavior changes. The KS values increase for some models, showing that the fit is not as good as before. The variation with aperture radius is also more visible. Some models perform better only in certain ranges, while others deviate more clearly.

5.4.3 Effect of Aperture Radius

For both beam configurations, the KS value is not constant and changes with aperture radius. There are specific regions where the KS value becomes smaller, indicating better agreement. Outside these regions, the error increases again.

5.4.4 Overall Observation

From all cases, it is clear that no single model works best for all conditions. The accuracy depends on both turbulence strength and aperture radius. More flexible models tend

to perform better over a wider range, while simpler models show larger deviations as turbulence increases.

6 Conclusion

In this study, the transmission of squeezed light through atmospheric channels was examined using different models of transmittance. The behavior of the probability distribution of transmittance (PDT) was studied under different turbulence conditions and beam configurations. It was observed that turbulence significantly affects the shape of the distribution. Under weak turbulence, the transmission remains relatively stable, while moderate and strong turbulence lead to increased fluctuations and losses. The effect of beam focusing was also seen to improve transmission mainly in weaker conditions. The comparison of different models shows that their accuracy depends on the channel conditions. Simpler models perform reasonably well in weak turbulence, but their performance decreases as turbulence increases. More flexible models are able to follow the distribution better in more complex situations.

Overall, the results show that the choice of model plays an important role in describing atmospheric quantum channels under different conditions.

References

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